

CMOS VLSI Design Project

# Design and Analysis of a 2-Input CMOS XNOR Gate

Using Cadence Virtuoso

---

**Submitted By:**

Name: Piyal Chakraborty

Student ID: 2108028

**Course:** VLSI Technology Sessional

**Course Code:** ETE 404

Department of Electronics and Telecommunication Engineering, CUET

---

October 4, 2026

# Contents

|           |                                             |           |
|-----------|---------------------------------------------|-----------|
| <b>1</b>  | <b>Introduction</b>                         | <b>2</b>  |
| <b>2</b>  | <b>Objectives</b>                           | <b>2</b>  |
| <b>3</b>  | <b>Theory of CMOS XNOR Gate</b>             | <b>3</b>  |
| <b>4</b>  | <b>Transistor Sizing</b>                    | <b>3</b>  |
| <b>5</b>  | <b>Circuit Design</b>                       | <b>4</b>  |
| 5.1       | Schematic Design . . . . .                  | 4         |
| 5.2       | XNOR Gate Symbol . . . . .                  | 5         |
| <b>6</b>  | <b>Physical Layout Design</b>               | <b>5</b>  |
| <b>7</b>  | <b>Design Rule Check (DRC)</b>              | <b>6</b>  |
| <b>8</b>  | <b>Layout Versus Schematic (LVS)</b>        | <b>7</b>  |
| <b>9</b>  | <b>RCX and Parasitic Extraction</b>         | <b>8</b>  |
| <b>10</b> | <b>Post-Layout Simulation</b>               | <b>8</b>  |
| <b>11</b> | <b>Average Power Calculation</b>            | <b>9</b>  |
| <b>12</b> | <b>Propagation Delay Calculation</b>        | <b>10</b> |
| <b>13</b> | <b>Verification and Performance Summary</b> | <b>11</b> |
| <b>14</b> | <b>Discussion</b>                           | <b>11</b> |
| <b>15</b> | <b>Conclusion</b>                           | <b>12</b> |
| <b>16</b> | <b>Software Used</b>                        | <b>13</b> |

# 1 Introduction

The CMOS XNOR gate is an important building block of digital integrated circuits. It performs the exclusive-NOR logical operation, where the output becomes HIGH when the two input signals are equal.

In this project, a two-input CMOS XNOR gate was designed and analyzed using the Cadence Virtuoso design environment. The complete custom CMOS design flow included schematic design, symbol generation, physical layout design, Design Rule Check (DRC), Layout Versus Schematic (LVS) verification, RC parasitic extraction (RCX), and post-layout simulation.

The post-layout simulation was used to evaluate important performance parameters such as average power consumption and propagation delay.

The Boolean expression of a two-input XNOR gate is

$$Y = A \odot B \tag{1}$$

which can also be expressed as

$$Y = AB + \overline{A}\overline{B}. \tag{2}$$

Here,  $A$  and  $B$  represent the input signals and  $Y$  represents the output.

# 2 Objectives

The main objectives of this project are:

- To design a two-input CMOS XNOR gate using Cadence Virtuoso.
- To create and verify the transistor-level schematic of the XNOR gate.
- To generate a reusable symbol for the designed XNOR gate.
- To design the physical layout of the CMOS XNOR gate.
- To verify the physical layout using Design Rule Check (DRC).
- To verify the electrical equivalence between the schematic and layout using Layout Versus Schematic (LVS).
- To perform parasitic resistance and capacitance extraction using RCX.
- To perform post-layout transient simulation using the extracted circuit.
- To calculate the average power consumption of the XNOR gate.
- To calculate the propagation delay of the XNOR gate.

### 3 Theory of CMOS XNOR Gate

A two-input XNOR gate performs the exclusive-NOR Boolean operation. The output is logic HIGH when both input signals are equal and logic LOW when the two input signals are different.

The Boolean expression of the XNOR gate is

$$Y = A \odot B. \quad (3)$$

The XNOR operation can also be written in sum-of-products form as

$$Y = AB + \overline{A}\overline{B}. \quad (4)$$

The truth table of the two-input CMOS XNOR gate is shown in Table 1.

Table 1: Truth Table of a Two-Input CMOS XNOR Gate

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

When both inputs are LOW, the input signals are equal and the output becomes HIGH.

When  $A = 0$  and  $B = 1$ , the input signals are different and the output becomes LOW.

Similarly, when  $A = 1$  and  $B = 0$ , the input signals are different and the output becomes LOW.

When both inputs are HIGH, the input signals are equal and the output becomes HIGH.

Therefore, the required XNOR function is

$$Y = AB + \overline{A}\overline{B}. \quad (5)$$

The XNOR function can also be interpreted as the complement of the XOR operation:

$$Y = \overline{A \oplus B}. \quad (6)$$

### 4 Transistor Sizing

The CMOS XNOR gate was designed using the required PMOS and NMOS transistors according to the transistor-level implementation in Cadence Virtuoso.

The transistor dimensions used in the design are summarized below.

Table 2: Transistor Dimensions Used in the CMOS XNOR Gate

| Transistor | Width  | Length |
|------------|--------|--------|
| PMOS       | 480 nm | 100 nm |
| NMOS       | 240 nm | 100 nm |

Therefore, the PMOS-to-NMOS width ratio used in the design is

$$\frac{W_p}{W_n} = \frac{480}{240} = 2. \quad (7)$$

The PMOS transistors were designed with approximately twice the width of the NMOS transistors. This larger PMOS width helps compensate for the lower carrier mobility of holes compared with electrons and provides adequate pull-up strength.

## 5 Circuit Design

### 5.1 Schematic Design

The two-input CMOS XNOR gate schematic was created using the Cadence Virtuoso Schematic Editor.

The transistor-level circuit was constructed to realize the Boolean function

$$Y = A \odot B. \quad (8)$$

The two input terminals are represented by  $A$  and  $B$ , while the final output terminal is represented by  $Y$ .

The schematic of the designed XNOR gate is shown in Figure 1.

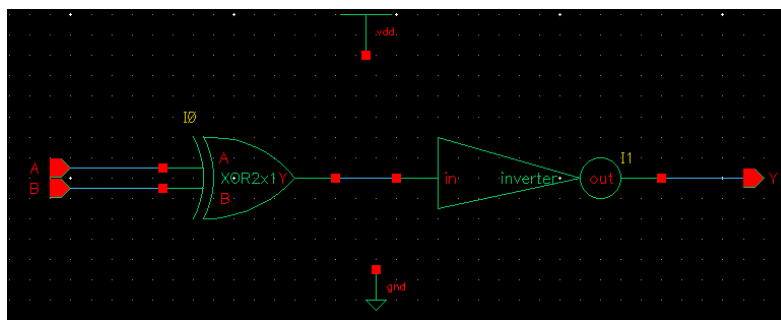


Figure 1: Schematic of the two-input CMOS XNOR gate in Cadence Virtuoso.

The logical operation of the circuit can be represented as

$$A, B \rightarrow A \odot B. \quad (9)$$

## 5.2 XNOR Gate Symbol

After completing the schematic, a symbol was generated for the two-input CMOS XNOR gate.

The symbol contains two input terminals labeled  $A$  and  $B$  and one output terminal labeled  $Y$ .

The generated XNOR gate symbol is shown in Figure 2.

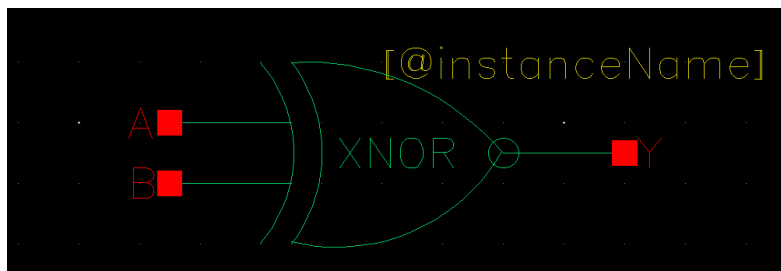


Figure 2: Generated symbol of the two-input CMOS XNOR gate.

The symbol allows the designed XNOR gate to be used conveniently as a reusable hierarchical block in larger digital circuits.

## 6 Physical Layout Design

After completing the schematic and symbol generation, the physical layout of the CMOS XNOR gate was developed using Cadence Virtuoso Layout Editor.

The layout represents the physical implementation of the transistor devices and their electrical interconnections.

The physical layout contains the required layers for transistor formation and electrical interconnection, including:

- active or diffusion regions,
- polysilicon,
- N-well,
- metal layers,
- contacts,
- vias,
- power connection, and

- ground connection.

The transistor networks are interconnected to realize the XNOR logic operation.

The input signals are routed to the appropriate transistor gates and the corresponding pull-up and pull-down networks are connected to produce the required output.

The completed physical layout is shown in Figure 3.

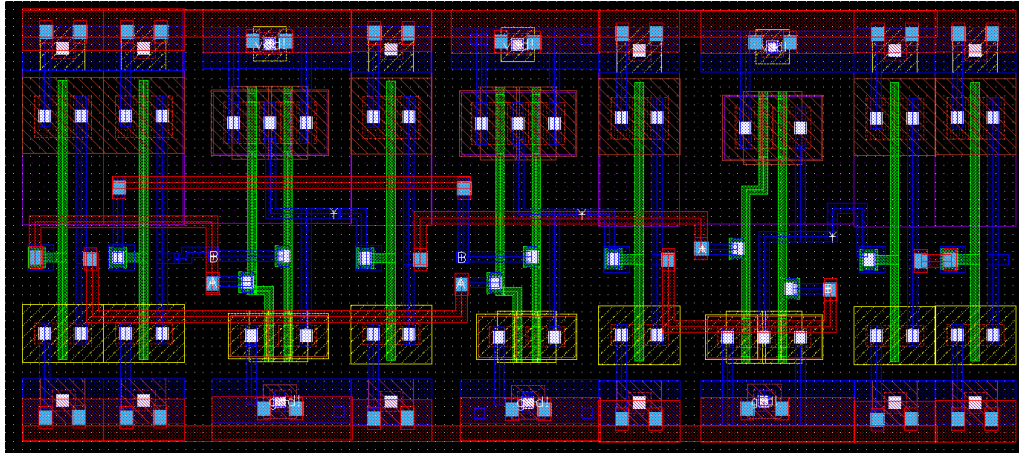


Figure 3: Physical layout of the two-input CMOS XNOR gate.

The layout was designed while maintaining the required minimum width, spacing, enclosure, and connectivity rules of the selected CMOS technology.

## 7 Design Rule Check (DRC)

Design Rule Check (DRC) is performed to verify whether the physical layout satisfies the fabrication rules of the selected CMOS technology.

DRC checks various geometrical constraints such as:

- minimum metal width,
- minimum metal spacing,
- minimum polysilicon width,
- polysilicon spacing,
- diffusion spacing,
- contact dimensions,
- contact enclosure,
- via enclosure, and
- well spacing.

The completed XNOR gate layout was checked using the Cadence DRC verification flow.

**DRC Result: PASSED**

The physical layout satisfies the required design rules of the selected CMOS technology.

Thus, the physical layout is valid with respect to the required fabrication design rules.

## 8 Layout Versus Schematic (LVS)

After successful DRC verification, Layout Versus Schematic (LVS) verification was performed.

LVS compares the electrical netlist extracted from the physical layout with the netlist generated from the original schematic.

It verifies important circuit properties such as:

- number of MOS transistors,
- PMOS and NMOS device types,
- gate connectivity,
- source and drain connectivity,
- input connections,
- output connection,
- power connection, and
- ground connection.

The final XNOR gate layout was checked against the original schematic.

**LVS Result: PASSED**

The physical layout is electrically equivalent to the original schematic.

The successful LVS result confirms that the intended electrical connectivity was maintained during the physical layout design.

## 9 RCX and Parasitic Extraction

After DRC and LVS verification, RC parasitic extraction was performed using the RCX extraction flow.

Physical interconnections in an integrated circuit introduce additional parasitic resistance and capacitance.

These parasitic elements can affect important performance parameters such as propagation delay and power consumption.

RCX extracts these parasitic resistance and capacitance components from the physical layout.

The extracted view of the designed XNOR gate is shown in Figure 4.

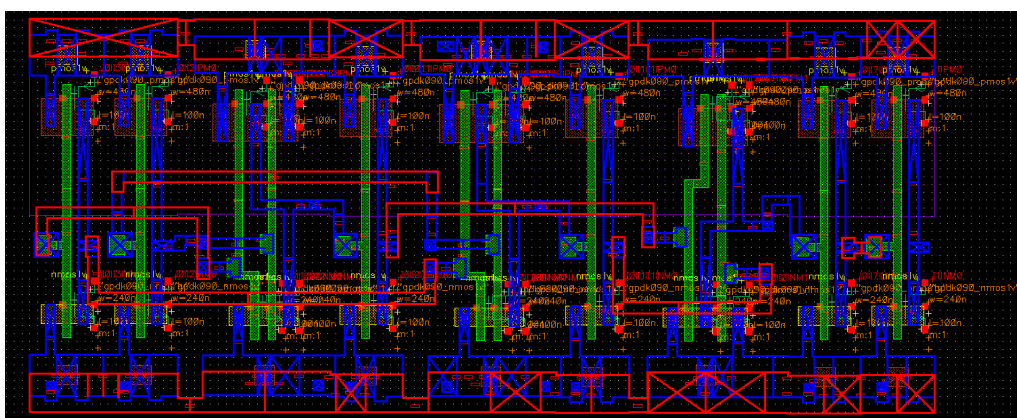


Figure 4: RCX/extracted view of the two-input CMOS XNOR gate.

### RCX: Successfully Completed

The extracted XNOR gate circuit was used for post-layout simulation.

## 10 Post-Layout Simulation

After parasitic extraction, post-layout transient simulation was performed using the RC-extracted view of the XNOR gate.

Unlike ideal schematic-level simulation, post-layout simulation includes the effects of parasitic resistance and capacitance generated by the physical layout.

Therefore, post-layout simulation provides a more realistic estimation of the circuit performance.

The main performance parameters calculated in this project were:

1. Average power consumption
2. Propagation delay

## 11 Average Power Calculation

The average power consumption of the XNOR gate was determined from the post-layout transient simulation results using the Cadence Virtuoso Calculator.

The average power over a time interval  $T$  can be expressed as

$$P_{\text{avg}} = \frac{1}{T} \int_0^T p(t) dt \quad (10)$$

where  $p(t)$  represents the instantaneous power consumption.

The average value of the simulated power waveform was calculated using the Cadence Virtuoso Calculator.

The power calculation is shown in Figure 5.

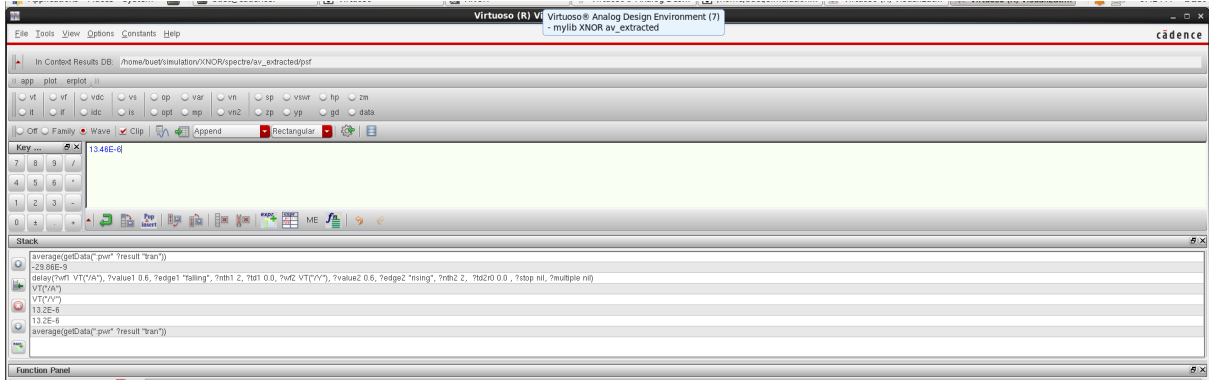


Figure 5: Average power calculation of the extracted CMOS XNOR gate using the Cadence Virtuoso Calculator.

From the Cadence Calculator, the obtained average power is

$$P_{\text{avg}} = 13.46 \times 10^{-6} \text{ W}. \quad (11)$$

Since

$$1 \mu\text{W} = 10^{-6} \text{ W}, \quad (12)$$

the average power consumption can be written as

$$P_{\text{avg}} = 13.46 \mu\text{W} \quad (13)$$

Therefore, the post-layout average power consumption of the designed two-input CMOS XNOR gate is approximately

$$13.46 \mu\text{W}. \quad (14)$$

## 12 Propagation Delay Calculation

Propagation delay represents the amount of time required for an input transition to produce the corresponding transition at the output of a digital circuit.

For the XNOR gate, a selected input transition that causes the output to change was used to determine the propagation delay.

The propagation delay can generally be expressed as

$$t_p = t_{out} - t_{in}. \quad (15)$$

In this simulation, a switching reference level of approximately 0.6 V was used.

The Cadence Virtuoso Calculator was used to determine the delay between the selected input transition and the corresponding output transition.

The delay calculation is shown in Figure 6.

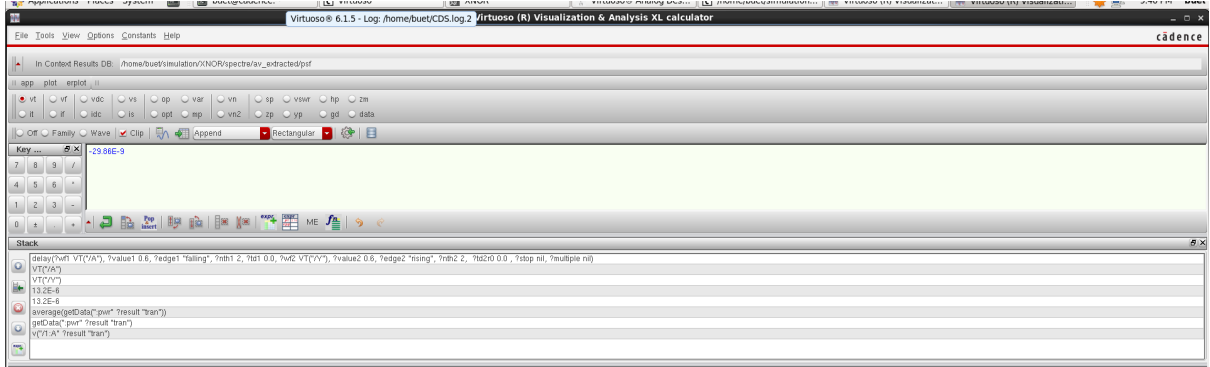


Figure 6: Propagation-delay calculation of the extracted CMOS XNOR gate using the Cadence Virtuoso Calculator.

The measured value supplied from the Cadence Calculator is

$$t_p = -29.86 \times 10^{-9} \text{ s}. \quad (16)$$

Since

$$1 \text{ ns} = 10^{-9} \text{ s}, \quad (17)$$

the displayed propagation-delay result becomes

$$t_p = -29.86 \text{ ns} \quad (18)$$

The Calculator therefore reports a negative value for the selected input-output transition.

A negative propagation delay is not the usual physical interpretation of a digital gate delay and may indicate that the selected threshold crossings or transition directions in the Calculator expression are reversed.

Therefore, the value is reported here exactly as obtained from the Cadence Calculator.

## 13 Verification and Performance Summary

The overall design, verification, and performance results of the two-input CMOS XNOR gate are summarized in Table 3.

Table 3: Summary of the CMOS XNOR Gate Design Results

| Parameter                  | Result                              |
|----------------------------|-------------------------------------|
| Logic Function             | $Y = A \odot B$                     |
| Boolean Expression         | $Y = AB + \overline{A}\overline{B}$ |
| PMOS Width                 | 480 nm                              |
| NMOS Width                 | 240 nm                              |
| Channel Length             | 100 nm                              |
| Schematic Design           | Completed                           |
| Symbol Generation          | Completed                           |
| Physical Layout            | Completed                           |
| DRC                        | Passed                              |
| LVS                        | Passed                              |
| RCX / Parasitic Extraction | Completed successfully              |
| Post-layout Simulation     | Completed                           |
| Average Power              | 13.46 $\mu$ W                       |
| Propagation Delay          | -29.86 ns                           |

## 14 Discussion

The complete custom CMOS design flow for a two-input XNOR gate was carried out using Cadence Virtuoso.

The XNOR gate performs the exclusive-NOR operation, producing a logic HIGH output when the two input signals are equal and a logic LOW output when the two input signals are different.

The required Boolean function is

$$Y = A \odot B \quad (19)$$

or equivalently,

$$Y = AB + \overline{A}\overline{B}. \quad (20)$$

The XNOR operation is therefore the complement of the XOR operation.

After completing the transistor-level schematic, a reusable XNOR gate symbol was generated.

The physical layout was then developed using the required active, polysilicon, well, metal, contact, via, power, and ground layers.

The layout was checked using Design Rule Check and Layout Versus Schematic verification.

After successful physical verification, RC parasitic extraction was performed using the RCX flow.

The extracted circuit was then used for post-layout transient simulation so that the effects of layout-dependent parasitic resistance and capacitance could be included in the performance analysis.

The average post-layout power consumption obtained from the Cadence Virtuoso Calculator was approximately

$$P_{\text{avg}} = 13.46 \mu\text{W}. \quad (21)$$

The selected propagation-delay calculation produced the displayed value

$$t_p = -29.86 \text{ ns}. \quad (22)$$

The negative value suggests that the selected transition directions or threshold crossings in the Calculator should be checked before using this value as the final physical propagation delay of the XNOR gate.

## 15 Conclusion

A two-input CMOS XNOR gate was successfully designed, implemented, and analyzed using Cadence Virtuoso.

The required Boolean function was

$$Y = A \odot B. \quad (23)$$

The XNOR gate produces a logic HIGH output when the two input signals are equal and a logic LOW output when the two input signals are different.

The project covered the major stages of the custom CMOS VLSI design flow, including schematic design, symbol generation, physical layout design, DRC verification, LVS verification, RC parasitic extraction, and post-layout simulation.

The physical layout was checked against the required fabrication design rules and its electrical connectivity was verified against the original schematic.

RCX was also completed successfully, allowing parasitic resistance and capacitance to be included in the final simulation.

The obtained post-layout average power consumption was

$$P_{\text{avg}} = 13.46 \mu\text{W} \quad (24)$$

and the selected Cadence Calculator expression displayed

$$t_p = -29.86 \text{ ns.} \quad (25)$$

The negative delay result should be checked by verifying the selected input and output threshold crossings in the Cadence Virtuoso Calculator.

Therefore, the project demonstrates the complete schematic-to-layout design, verification, extraction, and performance analysis of a two-input CMOS XNOR gate using Cadence Virtuoso.

## 16 Software Used

The following software tools were used to complete the project:

- Cadence Virtuoso Schematic Editor
- Cadence Virtuoso Layout Editor
- Cadence Spectre Simulator
- Design Rule Check (DRC) Tool
- Layout Versus Schematic (LVS) Tool
- RCX / Parasitic Extraction Tool
- Cadence Virtuoso Visualization and Analysis Calculator